STAT 385 Homework 5 Fall 2026

Due: Oct. 26, 2026

Background & Overview: This assignment compares variable selection, shrinkage, and dimension reduction for regression. Use R to fit models, select tuning parameters, and evaluate prediction accuracy. Submit your answers, clearly labeled R code, relevant output, and plots. Explain your conclusions in context.

Reproducibility and evaluation. Use the seeds specified below. In Q2 and Q3, sample `round(0.7*n)` observations without replacement for training and use the remaining observations for testing. Tune models using only the training data and 10-fold cross-validation (CV). Use the same folds across methods within each question. Estimate preprocessing parameters within each training fold. Keep the test set untouched until final evaluation.

Report test mean squared error (MSE) and root mean squared error (RMSE):

$$\text{MSE} = \frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} (y_i - \widehat{y}_i)^2, \qquad \text{RMSE} = \sqrt{\text{MSE}}.$$

Use `lambda.min` for ridge and lasso. Choose the number of PCR/PLS components to minimize CV MSE. Count nonzero predictor coefficients excluding the intercept.

Questions

Q1. Simulated Data and Variable Selection (43 points)

(a) Set `set.seed(2024)`. Generate $n = 100$ independent predictors $X_i \sim N(0, 1)$ and independent errors $\varepsilon_i \sim N(0, 0.25^2)$, independent of the predictors. Generate the entire predictor vector first, followed by the error vector. In `rnorm()`, 0.25 is the error standard deviation. [3 points]

(b) Generate $Y_i = 1 + 3X_i + 2X_i^2 - X_i^4 + \varepsilon_i$. [3 points]

(c) Create a data frame containing Y and the ten *raw powers* $X, X^2, \ldots, X^{10}$. Use `regsubsets()` from the `leaps` package with `nvmax=10` and exhaustive search. Among the best models of sizes 1 through 10, select models separately by minimum Mallows' C_p, minimum BIC, and maximum adjusted R^2. Plot each criterion versus model size, mark its optimum, and report the selected powers and coefficients for each criterion. Include an intercept in every model. [8 points]

(d) Repeat (c) using forward selection and backward selection. For each method and criterion, report model size, selected powers, and coefficient estimates. Compare with exhaustive search. [16 points]

(e) Fit lasso using the same ten powers, with standardized predictors and an unpenalized intercept. Set `set.seed(2024)` immediately before creating balanced random 10-fold assignments. Use CV to choose λ, plot CV MSE against $\log(\lambda)$, and report $\lambda_{\min}$, the intercept, and all nonzero coefficients on the original predictor scale. Does lasso recover the powers in the generating model? Explain. [5 points]

(f) Reuse the *same* X and ε from (a), but generate $Y_i = 1 + X_i^5 + \varepsilon_i$. Repeat exhaustive best subset selection under all three criteria and lasso with the *same folds* as in (e).

Report selected powers, coefficients, criterion plots, and the lasso CV plot. Compare with the true model. [8 points]

Q2. Predicting College Applications (29 points) Use `ISLR2::College`, with `Apps` as the response and all other variables as candidate predictors. Encode `Private` as a factor or a single indicator.

(a) Set `set.seed(2024)` and create the 70%/30% training/test split described above. Set `set.seed(2024)` again before generating training CV folds. Report the training and test sizes. [2 points]

(b) Fit ordinary least squares on all predictors using the training set. Report test MSE and RMSE. [3 points]

(c) Fit ridge regression, choosing λ by training-set CV. Report λ, test MSE, and test RMSE. [3 points]

(d) Repeat (c) for lasso and report the number of nonzero predictor coefficients. [5 points]

(e) Fit principal components regression (PCR) using standardized predictors. Choose M from $0, 1, \ldots, p$ by training-set CV, where p is the number of encoded predictors and $M = 0$ is the intercept-only model. Report the selected M and test errors. [6 points]

(f) Repeat (e) using partial least squares (PLS). [6 points]

(g) Summarize the five methods in a table. Compare their errors with an intercept-only benchmark that predicts the training mean of `Apps`. Discuss prediction accuracy relative to that benchmark and the scale of the response. Do not conclude that a method is inaccurate solely because its RMSE is numerically large. [4 points]

Q3. Predicting Boston Crime Rates (31 points) Use `ISLR2::Boston`, with `crim` as the response and all remaining variables, including `medv`, as candidate predictors. Use this version consistently: it has 12 predictors.

(a) Set `set.seed(2024)` and create a 70%/30% split. Reset the seed to 2024 and generate common training CV folds. Compare at least exhaustive best subset selection, ridge, lasso, and PCR. For subset selection, use CV to select model size from $0, \ldots, p$; repeat the subset search within every training fold. Report selected tuning parameters, training CV MSE, and final test MSE/RMSE. You may also include PLS. [16 points]

(b) Recommend a method using *training CV error*, then assess the fixed recommendation on the test set. Justify your recommendation in terms of prediction, complexity, and the benchmark. If you explore elastic net, choose both α and λ using training CV; do not choose them by test error. [10 points]

(c) List the predictors used by your recommended model. Does it use all available predictors? Explain the distinction between selecting original variables and constructing components from them. [5 points]